Information Retrieval: BM25 and Dense Retrieval

Cynaris AI/ML Internship - Week 7 Reference Material

1. Information Retrieval

Information retrieval identifies and ranks documents that are useful for a user's information need. Search systems commonly return the top K matching documents.

2. BM25 Retrieval

BM25 is a sparse lexical ranking algorithm based on term frequency, inverse document frequency and document length normalization. It is effective when query terms overlap with relevant document terms.

3. BM25 Limitations

BM25 may fail when a query uses synonyms or different wording from the source document. It primarily measures lexical matching rather than semantic similarity.

4. Dense Retrieval

Dense retrieval converts documents and queries into numeric embedding vectors. A similarity function such as cosine similarity or dot product ranks documents according to their vector representations.

5. Embeddings

An embedding model maps text into a vector space. Texts with related meanings may have nearby vectors, even when they do not share the same exact words. Query and document embeddings must use compatible models.

6. Precision at K

Precision at K is the number of relevant retrieved documents among the first K results divided by K. For example, if 3 of the top 5 results are relevant, Precision@5 equals 0.60.

7. Recall at K

Recall at K is the number of relevant retrieved documents among the first K results divided by the total number of known relevant documents for that query.

8. Mean Reciprocal Rank

Mean Reciprocal Rank averages the reciprocal rank of the first relevant result across queries. It rewards systems that place a relevant result near the top.

9. Fair Comparison

BM25 and dense retrieval should use the same documents, the same ten queries, the same top K and the same human relevance judgments. Report per-query results as well as aggregate metrics.

10. Limitations

Human relevance judgments can be subjective. A small dataset may not represent real-world search behavior. Embedding quality depends on the selected model and the domain of the documents.